PROJECT DOCUMENTATION

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Agenda

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Highlight the text you want to hyperlink, select the link symbol, then select from the dropdown the page that you want to connect.

- Project Overview
- Architecture (Industry Standard)
- Flow Diagram
- Project Structure
- Database Design
- Authentication (JWT)
- JWT Configuration
- Key Features Implemented
- References
- Whiteboard
- Brainstorm

1. Project Overview

This project is a Customer Support Ticketing System API built using:

- .NET 10 Web API
- Entity Framework Core
- SQL Server
- JWT Authentication
- Clean Architecture (Repository + Service Pattern)

Purpose:

To manage user support tickets with authentication and secure APIs.

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2. Architecture (Industry Standard)

Controllers → API Layer
Services → Business Logic
Repositories → Data Access Layer
DTOs → Data Transfer
Helpers → Utility (JWT, Middleware)

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Flow Diagram

Client → Controller → Service → Repository → Database

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3. Project Structure

```
TicketingSystem/
│
├─── Controllers/
├─── Services/
├─── Services/Interfaces/
├─── Repositories/
├─── Repositories/Interfaces/
├─── Models/
├─── DTOs/
├─── Data/
├─── Helpers/
├─── Migrations/
└─── Program.cs
```

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4. Database Design

User Table

Column	Type	Description
Id	int (PK)	User ID
Email	string	Unique email
Password	string	Hashed password
Role	string	Admin/User
CreatedAt	datetime	Created time

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5. Authentication (JWT)

Register → Save User
Login → Generate JWT Token
Token → Access Protected APIs

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JWT Configuration

```
{
  "Jwt": {
    "Key": "your-secret-key",
    "Issuer": "your-app",
    "Audience": "your-app-users"
  }
}
```

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6. Key Features Implemented

User Registration
User Login (JWT)
Password Hashing (BCrypt)
Dependency Injection
Global Exception Handling
Swagger Integration
SQL Server Integration
Auto Table Creation (EF Core)

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Dependencie Documentation

This document describes all NuGet packages used in the TicketingSystem API, including their purpose, usage, and importance.

Which libraries did you use?

I used JWT Bearer for authentication, BCrypt for password hashing, Entity Framework Core with SQL Server for ORM, and Swagger for API documentation, all aligned with .NET 10.

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1. Authentication & Security

BCrypt.Net-Next (v4.1.0)

Purpose: Password hashing

Usage:

- Securely hash user passwords during registration
- Verify password during login

Why Used:

- Prevents storing plain text passwords
- Automatically handles salting

Example:

```
var hash = BCrypt.Net.BCrypt.HashPassword(password);
var isValid = BCrypt.Net.BCrypt.Verify(password, hash);
```

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Microsoft.AspNetCore.Authentication.JwtBearer (v10.0.5)

Purpose: JWT Authentication

Usage:

- Generate and validate JWT tokens
- Secure API endpoints

Why Used:

- Stateless authentication
- Industry standard for APIs

Example:

```
builder.Services.AddAuthentication(JwtBearerDefaults.Authentication
Scheme)
    .AddJwtBearer();
```

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2. Framework & API Core

Microsoft.AspNetCore.OpenApi (v10.0.5)

Purpose: OpenAPI support

Usage:

- Enables API metadata generation
- Works with Swagger

Why Used:

- Standard API documentation format
- Improves API discoverability

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3. Database & ORM

Microsoft.EntityFrameworkCore.SqlServer (v10.0.5)

Purpose: SQL Server database provider

Usage:

- Connect API with SQL Server
- Perform CRUD operations

Why Used:

- Strong integration with .NET
- Supports migrations and LINQ queries

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Microsoft.EntityFrameworkCore.Design (v10.0.5)

Purpose: Development-time tools

Usage:

- Run migrations
- Scaffold database

Command Examples:

dotnet ef migrations add InitialCreate

dotnet ef database update

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4. API Documentation

Swashbuckle.AspNetCore (v10.1.7)

Purpose: Swagger UI

Usage:

- Test APIs in browser
- View endpoints and models

Why Used:

- Interactive API testing
- Auto-generated documentation

Access URL:

https://localhost:xxxx/swagger

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